Packet Tracer - Configure AAA Authentication on Cisco Routers

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/0/1	192.168.1.1	255.255.255.0	N/A	S1 G1/0/1
	S0/1/0	10.1.1.2	255.255.255.252		N/A
R2	G0/0/0	192.168.2.1	255.255.255.0	N/A	S2 G1/0/1
	S0/1/0	10.1.1.1	255.255.255.252		N/A
	S0/1/1	10.2.2.1	255.255.255.252		N/A
R3	G0/0/1	192.168.3.1	255.255.255.0	N/A	S3 G1/0/1
	S0/1/1	10.2.2.2	255.255.255.252		N/A
TACACS+ Server	NIC	192.168.2.2	255.255.255.0	192.168.2.1	S2 G1/0/24
RADIUS Server	NIC	192.168.3.2	255.255.255.0	192.168.3.1	S3 G1/0/24
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 G1/0/24
PC-B	NIC	192.168.2.3	255.255.255.0	192.168.2.1	S2 G1/0/23
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 G1/0/23

Objectives

- Configure a local user account on R1 and configure authenticate on the console and vty lines using local AAA.
- Verify local AAA authentication from the R1 console and the PC-A client.
- Configure server-based AAA authentication using TACACS+.
- Verify server-based AAA authentication from the PC-B client.
- Configure server-based AAA authentication using RADIUS.
- Verify server-based AAA authentication from the PC-C client.

Background / Scenario

The network topology shows routers R1, R2 and R3. Currently, all administrative security is based on knowledge of the enable secret password. Your task is to configure and test local and server-based AAA solutions.

You will create a local user account and configure local AAA on router R1 to test the console and vty logins. You will then configure router R2 to support server-based authentication using the TACACS+ protocol. Finally, you will configure router R3 to support server-based authentication using the RADIUS protocol.

The devices in the topology are initially configured with the following passwords:

- Console password: **ciscoconpa55**
- Enable secret password: **ciscoenpa55**

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Note: IOS version 15.3 uses SCRYPT as a secure encryption hashing algorithm; however, the IOS version that is currently supported in Packet Tracer uses MD5. Always use the most secure option available on your equipment.

Instructions

Part 1: Configure Local AAA Authentication for Console Access on R1

Step 1: Test connectivity.

- Ping from **PC-A** to **PC-B**.
- Ping from **PC-A** to **PC-C**.
- Ping from **PC-B** to **PC-C**.

Step 2: Configure a local username on R1.

Configure a username of **Admin1** with a secret password of **admin1pa55**.

Step 3: Configure local AAA authentication for console access on R1.

Enable AAA on R1 and configure AAA authentication for the console login to use the local database.

Step 4: Configure the line console to use the defined AAA authentication method.

Enable AAA on **R1** and configure AAA authentication for the console login to use the default method list.

Step 5: Verify the AAA authentication method.

Verify the user EXEC login using the local database.

Part 2: Configure Local AAA Authentication for vty Lines on R1

Step 1: Configure domain name and crypto key for use with SSH.

a. Use ccnp.com as the domain name on R1.

b. Create an RSA crypto key using 2048 bits.

Step 2: Configure a named list AAA authentication method for the vty lines on R1.

Configure a named list called **SSH-LOGIN** to authenticate logins using local AAA.

Step 3: Configure the vty lines to use the defined AAA authentication method.

Configure the vty lines to use the named AAA method and only allow SSH for remote access.

Step 4: Verify the AAA authentication method.

Verify the SSH configuration SSH to **R1** from the command prompt of **PC-A**.

Part 3: Configure Server-Based AAA Authentication Using TACACS+ on R2

Step 1: Configure a backup local database entry called Admin.

For backup purposes, configure a local username of **Admin2** and a secret password of **admin2pa55**.

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Step 2: Verify the TACACS+ Server configuration.

a. Ping the TACACS+ Server to verify that it is reachable from R2.

b. Click the TACACS+ Server. On the Services tab, click **AAA**. Notice that there is a Network configuration entry for **R2** and a User Setup entry for **Admin2**.

Step 3: Configure the TACACS+ server specifics on R2.

Configure the AAA TACACS server IP address and secret key on **R2**.

Note: The commands **tacacs-server host** and **tacacs-server key** are deprecated. Currently, Packet Tracer does not support the new command **tacacs server**.

```
R2(config)# tacacs-server host 192.168.2.2
R2(config)# tacacs-server key tacacspa55
```

Step 4: Configure AAA login authentication for console access on R2.

Enable AAA on **R2** and configure all logins to authenticate using the AAA TACACS+ server. If it is not available, then use the local database.

Step 5: Configure the line console to use the defined AAA authentication method.

Configure AAA authentication for console login to use the default AAA authentication method.

Step 6: Verify the AAA authentication method.

Verify the user EXEC login using the AAA TACACS+ server.

Part 4: Configure Server-Based AAA Authentication Using RADIUS on R3

Step 1: Configure a backup local database entry called Admin3.

For backup purposes, configure a local username of **Admin3** and a secret password of **admin3pa55**.

Step 2: Verify the RADIUS Server configuration.

Click the RADIUS Server. On the Services tab, click **AAA**. Notice that there is a Network configuration entry for **R3** and a User Setup entry for **Admin3**.

Step 3: Configure AAA login authentication for console access on R3.

Enable AAA on **R3** and configure all logins to authenticate using the AAA RADIUS server. If it is not available, then use the local database.

Step 4: Configure the RADIUS server specifics on R3.

Configure the AAA RADIUS server IP address and secret key on **R3**. Configure the server name as **RAD-SERVER**.

Note: Packet Tracer does not correctly score configuration using the **radius server** command.

Step 5: Configure the line console to use the defined AAA authentication method.

Configure AAA authentication for console login to use the default AAA authentication method.

Step 6: Verify the AAA authentication method.

Verify the user EXEC login using the AAA RADIUS server.

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Step 7: Check results.

Your completion percentage should be 100%. Click **Check Results** to see feedback and verification of which required components have been completed.

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